Complementarity Problem Types

Special Case: Single-valued complementarity

S.I.C.P(f, g, K): Simultaneous Implicit
$x_0 \in K$: $f(x_0) \in K^*$, $g(x_0) \in K^*$

S.E.C.P(f, g, K): Simultaneous Explicit
$x_0 \in K$: $f(x_0) \in K^*$, $g(x_0) \in K^*$

I.C.P: Implicit Complementarity
$x_0 \in K$: $f(x_0) \in K^*$, $g(x_0) \in K$, $\langle g(x_0), f(x_0) \rangle = 0$

E.C.P: Explicit Complementarity
$x_0 \in K$: $f(x_0) \in K^*$, $\langle x_0, f(x_0) \rangle = 0$